LOGARITHMS

Common or natural?

The choice depends on context.

USE common logarithms FOR

SCALES	Orders of magnitude and human-readable scales: the Richter scale, the decibel scale, the pH scale all use base 10.
DECIMAL DATA	Decimal-based measurements; a change of 1 on the Richter and pH scales is a factor of 10.

USE natural logarithms FOR

GROWTH AND DECAY	Continuous growth or decay: models of the form Ae^{kt} pair naturally with $\ln$.
CALCULUS	Calculus and theoretical derivations.
BASE e EQUATIONS	Solving $e^{2x} = 5$: $2x = \ln(5)$, so $x = \ln(5)/2$.

EITHER WORKS

$3^x = 7$ gives $x = \log(7)/\log(3)$ or $x = \ln(7)/\ln(3)$, the same number; divide the two logarithms as numbers, not as $\log(7/3)$.